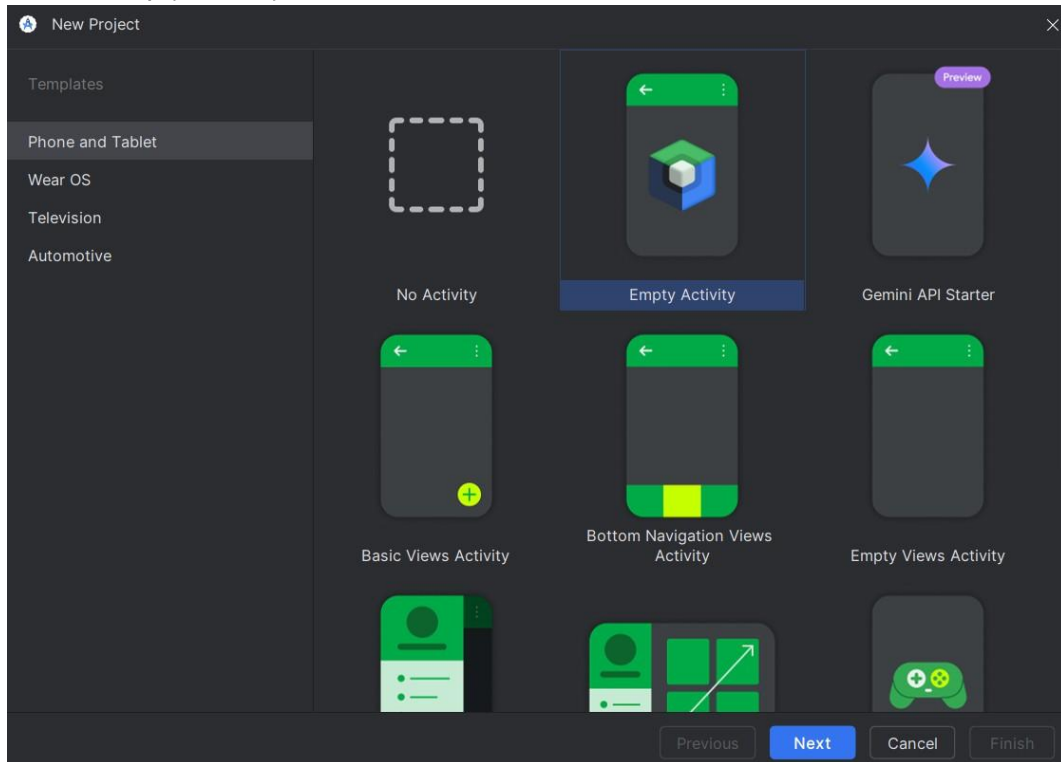


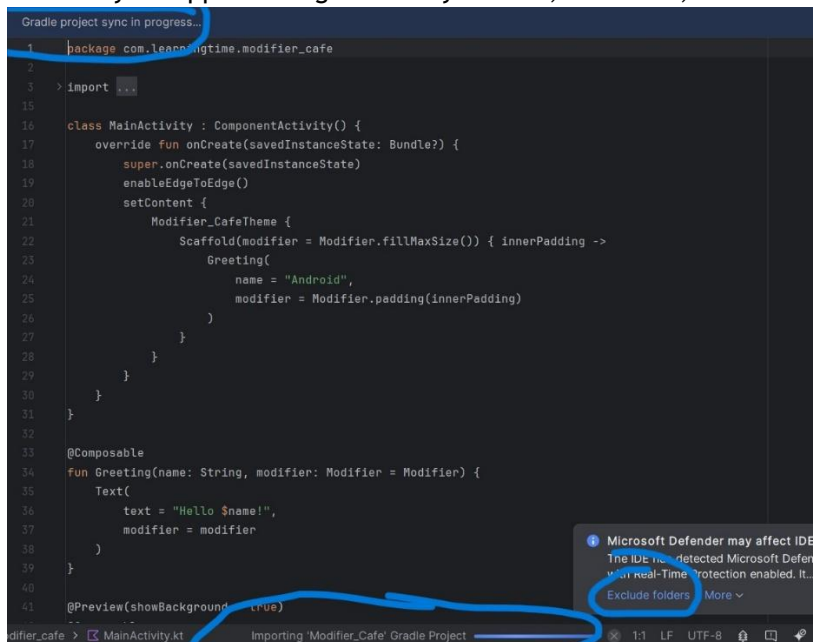
UI CONTROLS - III

ANDROID APPLICATION DEVELOPMENT - 20CYS404

1. Open Android Studio and Create a new Project
2. Click on Empty Activity



3. Write a name for your Application and then click Finish.
4. Wait for your application gradle to sync. Also, if needed, click the exclude folders option.



5. After the sync, your application will be available to run.
6. Finally, write this code in your MainActivity.kt file. The comments explain the code for your understanding.

```
package com.learningtime.modifier_cafe

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.clickable
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material.icons.Icons
import androidx.compose.material.icons.filled.ShoppingCart
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.tooling.preview.Preview
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import com.learningtime.modifier_cafe.ui.theme.Modifier_CafeTheme

/**
 * MainActivity - Entry point of the Android application
 * This is the default activity that launches when the app starts
 */
class MainActivity : ComponentActivity() {
```

```

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)

    // Set the Compose UI as the content view
    setContent {
        // Apply the app's theme
        Modifier_CafeTheme {
            // Display the main composable function
            ModifierCafeApp()
        }
    }
}

```

/**

* OPTIMIZED EXPLANATION:

* The Modifier Cafe app simulates a coffee shop interface where users can:

* 1. Select coffee sizes (Small, Medium, Large)

* 2. Change coffee types by tapping the coffee cup (Latte, Cappuccino, Espresso)

* 3. Place orders using the FAB (Floating Action Button)

* 4. Track order count in real-time

*

* The app demonstrates:

* - State management with remember/mutableStateOf

* - Layout composition with Column, Row, Box

* - Material 3 components (Card, Surface, FAB)

* - Modifier usage for styling and positioning

*/

```
@OptIn(ExperimentalMaterial3Api::class) // Required for Material 3 features
```

```
@Composable
```

```
fun ModifierCafeApp() {
```

```
    // ===== STATE MANAGEMENT =====
```

```
    // State variables that trigger recomposition when changed
```

```
    // Tracks the selected coffee size
```

```
    var coffeeSize by remember { mutableStateOf("Medium") }
```

```
    // Tracks the selected coffee type
```

```
    var coffeeType by remember { mutableStateOf("Latte") }
```

```
    // Tracks the total number of orders placed
```

```
    var orderCount by remember { mutableStateOf(0) }
```

```
    // ===== MAIN CONTAINER =====
```

```
    // Background surface for the entire app
```

```
    Surface(
```

```
        modifier = Modifier.fillMaxSize(), // Takes up all available space
```

```
        color = Color(0xFFFF5F0E1) // Light beige color resembling cafe walls
```

```
    ) {
```

```
        // ===== PRIMARY LAYOUT =====
```

```
        // Vertical column that holds all UI elements
```

```
        Column(
```

```
            modifier = Modifier
```

```
                .fillMaxSize() // Take full available space
```

```
                .systemBarsPadding() // Avoid system UI (status bar, navigation bar)
```

```
                .padding(16.dp) // Add padding inside the column
```

```
        ) {
```

```
// ===== TITLE CARD =====
```

```
// Header section with app title
```

```
Card(
```

```
    modifier = Modifier
```

```
        .fillMaxWidth() // Stretch to full width
```

```
        .height(60.dp), // Fixed height
```

```
    colors = CardDefaults.cardColors(
```

```
        containerColor = Color(0xFF795548) // Brown color
```

```
    )
```

```
) {
```

```
    // Centered box for the title text
```

```
    Box(
```

```
        modifier = Modifier.fillMaxSize(),
```

```
        contentAlignment = Alignment.Center
```

```
    ) {
```

```
        Text(
```

```
            text = "Modifier Café", // App title
```

```
            fontSize = 24.sp,
```

```
            fontWeight = FontWeight.Bold,
```

```
            color = Color.White
```

```
        )
```

```
    }
```

```
}
```

```
Spacer(modifier = Modifier.height(16.dp)) // Vertical spacing
```

```
// ===== SIZE SELECTOR SECTION =====
```

```
// Label for size selection
```

```
Text(
```

```
text = "Select Size:",  
fontSize = 18.sp,  
fontWeight = FontWeight.Bold,  
modifier = Modifier.padding(bottom = 8.dp)  
)
```

```
// Horizontal row containing all size options
```

```
Row(modifier = Modifier.fillMaxWidth()) {
```

```
    // Small size option
```

```
    SizeOption(  
        size = "Small",  
        selectedSize = coffeeSize,  
        modifier = Modifier.weight(1f), // Takes equal space  
        onClick = { coffeeSize = "Small" }  
    )
```

```
    Spacer(modifier = Modifier.width(8.dp)) // Horizontal spacing
```

```
    // Medium size option
```

```
    SizeOption(  
        size = "Medium",  
        selectedSize = coffeeSize,  
        modifier = Modifier.weight(1f),  
        onClick = { coffeeSize = "Medium" }  
    )
```

```
    Spacer(modifier = Modifier.width(8.dp))
```

```
    // Large size option
```

```

SizeOption(
    size = "Large",
    selectedSize = coffeeSize,
    modifier = Modifier.weight(1f),
    onClick = { coffeeSize = "Large" }
)
}

```

```

Spacer(modifier = Modifier.height(24.dp)) // Vertical spacing

```

```

// ===== MAIN CONTENT AREA =====

```

```

// This area contains the interactive coffee table and order counter

```

```

Row(
    modifier = Modifier
        .fillMaxWidth() // Full width
        .weight(1f), // Takes remaining vertical space
    verticalAlignment = Alignment.CenterVertically
) {

```

```

// ===== LEFT SIDE - COFFEE TABLE =====

```

```

// Interactive coffee cup area

```

```

Box(
    modifier = Modifier
        .weight(1f) // Takes 50% of horizontal space
        .fillMaxHeight(),
    contentAlignment = Alignment.Center
) {

```

```

// Wooden table surface

```

```

Surface(
    modifier = Modifier.size(180.dp), // Fixed size

```

```

shape = RoundedRectangleBorder(16.dp), // Rounded corners
tonalElevation = 16.dp, // Shadow effect
color = Color(0xFF8B4513) // Brown wood color
) {
    // Container for the coffee cup card

    Box(
        modifier = Modifier.fillMaxSize(),
        contentAlignment = Alignment.Center
    ) {
        // Interactive coffee cup card

        Card(
            modifier = Modifier
                .size(120.dp) // Fixed size
                .clickable { // Makes it tappable
                    // Cycle through coffee types on tap
                    coffeeType = when (coffeeType) {
                        "Latte" -> "Cappuccino"
                        "Cappuccino" -> "Espresso"
                        else -> "Latte"
                    }
                },
            elevation = CardDefaults.cardElevation(defaultElevation = 8.dp)
        ) {
            // Content inside the coffee cup card

            Column(
                modifier = Modifier
                    .fillMaxSize()
                    .padding(8.dp),
                horizontalAlignment = Alignment.CenterHorizontally,

```

```

        verticalArrangement = Arrangement.Center
    ){
        Text(
            text = coffeeType, // Current coffee type
            fontSize = 14.sp,
            fontWeight = FontWeight.Bold,
            color = Color(0xFF5D4037) // Dark brown
        )
        Spacer(modifier = Modifier.height(2.dp))
        Text(
            text = "Tap", // Instruction text
            fontSize = 10.sp,
            color = Color.Gray
        )
    }
}
}
}
}
}
}

```

```

        Spacer(modifier = Modifier.height(50.dp)) // Vertical spacer (acts as visual
separation)

```

```

// ===== RIGHT SIDE - ORDER COUNTER =====

```

```

Column(
    modifier = Modifier
        .width(140.dp) // Fixed width
        .fillMaxHeight()
        .offset(x = 10.dp, y = 40.dp), // Slight offset for visual interest
    horizontalAlignment = Alignment.CenterHorizontally
)

```

```

){
    // ===== ORDER COUNTER CARD =====
    // Displays the total number of orders

    Card(
        modifier = Modifier.size(120.dp),
        colors = CardDefaults.cardColors(
            containerColor = Color(0xFFFF9800) // Orange color
        )
    ){
        Box(
            modifier = Modifier.fillMaxSize(),
            contentAlignment = Alignment.Center
        ){
            Column(horizontalAlignment = Alignment.CenterHorizontally) {
                Text(
                    text = "Orders",
                    fontSize = 14.sp,
                    color = Color.White
                )
                Text(
                    text = "$orderCount", // Dynamic order count
                    fontSize = 28.sp,
                    fontWeight = FontWeight.Bold,
                    color = Color.White
                )
                Text(
                    text = "Tap FAB →", // Instruction text
                    fontSize = 10.sp,
                    color = Color.White.copy(alpha = 0.8f) // Semi-transparent

```

```

    )
  }
}
}

```

```

Spacer(modifier = Modifier.height(12.dp))

```

```

// ===== INSTRUCTIONS CARD =====

```

```

// Provides user instructions

```

```

Card(

```

```

    modifier = Modifier.fillMaxWidth(),

```

```

    colors = CardDefaults.cardColors(

```

```

        containerColor = Color.White.copy(alpha = 0.9f) // Semi-transparent

```

white

```

    )

```

```

){

```

```

    Column(modifier = Modifier.padding(12.dp)) {

```

```

        Text(

```

```

            text = "Try:",

```

```

            fontSize = 14.sp,

```

```

            fontWeight = FontWeight.Bold,

```

```

            color = Color.Black

```

```

        )

```

```

        Spacer(modifier = Modifier.height(4.dp))

```

```

        // Step-by-step instructions

```

```

        Text("1. Tap cup", fontSize = 12.sp, color = Color.Black)

```

```

        Text("2. Select size", fontSize = 12.sp, color = Color.Black)

```

```

        Text("3. Click on FAB", fontSize = 12.sp, color = Color.Black)

```

```

    }

```

```

    }
  }
}

```

```

Spacer(modifier = Modifier.height(16.dp))

```

```

// ===== BOTTOM SECTION =====

```

```

// Contains current order info and the FAB

```

```

Box(

```

```

    modifier = Modifier

```

```

        .fillMaxWidth()

```

```

        .height(120.dp)

```

```

) {

```

```

    // ===== CURRENT ORDER CARD =====

```

```

    // Shows the current selection

```

```

    Card(

```

```

        modifier = Modifier

```

```

        .fillMaxWidth(0.7f) // Takes 70% of width

```

```

        .height(100.dp)

```

```

        .align(Alignment.CenterStart) // Align to left

```

```

        .padding(start = 16.dp),

```

```

        colors = CardDefaults.cardColors(

```

```

            containerColor = Color.White

```

```

        )

```

```

    ) {

```

```

        Row(

```

```

            modifier = Modifier

```

```

            .fillMaxSize()

```

```

            .padding(horizontal = 16.dp),

```

```

        verticalAlignment = Alignment.CenterVertically,
        horizontalArrangement = Arrangement.SpaceBetween
    ){
        Column {
            Text(
                text = "Current Order",
                fontSize = 12.sp,
                color = Color.Gray
            )
            Text(
                text = "$coffeeType - $coffeeSize", // Current selection
                fontSize = 16.sp,
                fontWeight = FontWeight.Bold,
                color = Color.Black
            )
            Text(
                text = "Orders: $orderCount", // Order count
                fontSize = 14.sp,
                color = Color(0xFF4CAF50), // Green color
                fontWeight = FontWeight.Bold
            )
        }
    }
}

```

// ===== FLOATING ACTION BUTTON (FAB) =====

// Primary action button to place orders

```

FloatingActionButton(
    modifier = Modifier

```



```

onClick: () -> Unit
){
    // Check if this size option is currently selected
    val isSelected = size == selectedSize

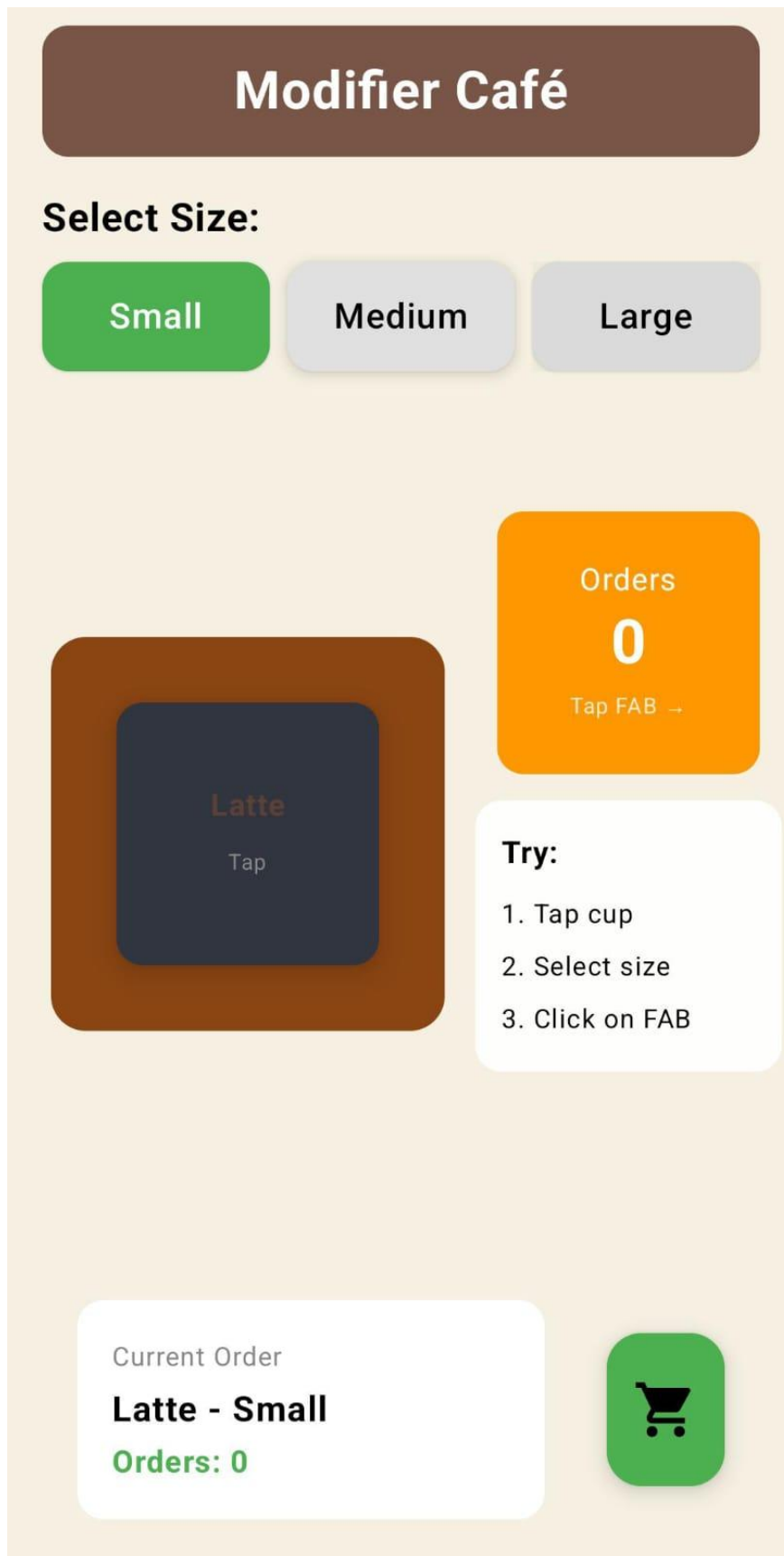
    // Card that acts as a clickable button
    Card(
        modifier = modifier
            .height(50.dp) // Fixed height
            .clickable { onClick() }, // Makes it tappable
        colors = CardDefaults.cardColors(
            // Green if selected, gray if not
            containerColor = if (isSelected) Color(0xFF4CAF50) else Color(0xFFE0E0E0)
        ),
        elevation = CardDefaults.cardElevation(
            // Higher elevation when selected for visual feedback
            defaultElevation = if (isSelected) 4.dp else 1.dp
        )
    ){
        // Centered text inside the card
        Box(
            modifier = Modifier.fillMaxSize(),
            contentAlignment = Alignment.Center
        ){
            Text(
                text = size,
                fontSize = 16.sp,
                fontWeight = FontWeight.Medium,
                // White text on green background when selected

```

```
        color = if (isSelected) Color.White else Color.Black
    )
}
}
}

/**
 * Preview function for Android Studio's Design tab
 * Allows developers to see the UI without running the app
 */
@Preview(showBackground = true)
@Composable
fun ModifierCafePreview() {
    Modifier_CafeTheme {
        ModifierCafeApp()
    }
}
```


Output:



Key Concepts:

1. State Management

- Uses `remember` and `mutableStateOf` for reactive UI updates
- Three state variables track: coffee size, coffee type, and order count

2. Layout Structure

- **Column:** Vertical arrangement of sections
- **Row:** Horizontal arrangement of size buttons and main content
- **Box:** For precise positioning (centering, overlapping, alignment)
- **Spacer:** For consistent spacing between elements

3. Modifier System

- `.fillMaxSize()`: Takes all available space
- `.padding()`: Adds internal spacing
- `.clickable()`: Makes elements interactive
- `.weight()`: Proportional space allocation
- `.offset()`: Manual positioning adjustments
- `.systemBarsPadding()`: Respects system UI boundaries

4. Material 3 Components

- **Card:** Container with elevation and rounded corners
- **Surface:** Base container with color and elevation
- **FloatingActionButton:** Prominent action button
- **Text:** Typography with styling options

5. Interactive Elements

- Coffee cup: Tap to cycle through coffee types
- Size buttons: Select different sizes with visual feedback
- FAB: Tap to increment order counter

6. Visual Design

- Cafe-themed color palette (browns, beige, green)
- Consistent spacing and alignment

- Visual feedback for selected states
- Clear hierarchy through typography and color